The in-plane thermal conductivity of lithium-ion cells: Parametric influences and simulative prediction

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***Abstract*—The in-plane thermal conductivity of lithium-ion cells is a crucial parameter for thermal management design of battery systems. However, only few measurement data and very limited information about parametric effects are available. Therefore, a simple but accurate measurement setup is presented which allows for the modification of parametric influences. Following, the effect of temperature, state of charge and state of health on the in-plane thermal conductivity is analyzed with tailored measurements for the first time. Finally, the commonly utilized approach of parallel thermal resistances is validated with experimental data first-ever.**

It is found that the in-plane thermal conductivity of pristine cells is nearly constant over all operating ranges. Neither temperature, nor state of charge significantly influence the in-plane thermal conductivity of lithium-ion cells. While aging does not drastically influence the in-plane thermal resistance of the cell, the cell thickness increase due to aging can significantly decrease the resulting in-plane thermal conductivity. As the experimental data matches the analytically calculated value which is obtained through parallel thermal resistances, it can be demonstrated that the prediction of the in-plane thermal conductivity with analytical calculations is accurate when cell layer thicknesses are known. Furthermore, it is proven that the copper and aluminum highly dominate the in-plane thermal conductivity.

***Index Terms*—Lithium-ion batteries, In-plane thermal conductivity, Thermal management design, Thermal resistance**

I. Introduction

Thermal management of lithium-ion batteries is crucial for ensuring long service life, preventing safety critical events and fulfilling high performance demands [1]. Besides heat capacity and through-plane thermal conductivity, the in-plane thermal conductivity is essential for predicting thermal behavior and ensuring the adherence to thermal limits. However, there only exists few experimental data for the in-plane thermal conductivity in literature. Furthermore, the scattering of the measured in-plane thermal conductivity is significant. Varying between 20 $Wm^{-1}K^{-1}$ to 40 $Wm^{-1}K^{-1}$ [2], selecting a distinct in-plane thermal conductivity value when parametrizing a model with satisfying accuracy is not possible. A common approach for analytically predicting the thermal conductivity is utilizing a thermal resistance network connected in parallel [3]–[5]. Though, this simulative calculation lacks experimental validation. Analytically obtained values have never been compared to experimentally measured in-plane thermal conductivities and thus the method is not reliably applicable.

In addition, there exist many different measurement approaches for characterizing the in-plane thermal conductivity of lithium-ion cells. While Aiello et al. [6] developed a thermoelectric device utilizing a temperature-guarded plate with heat flux sensors, Barreras et al. [7] employed a thermography test bench for measuring the thermal diffusivity. Additionally, Zhang et al. [8] measured the in-plane thermal conductivity by utilizing a circular heater between two sandwiched cells and iteratively adjusting thermal characteristics in a temperature simulation to fit measured temperature data. In order to measure the in-plane thermal conductivity of cylindrical cells, Drake et al. [9] and Spinner et al. [10] wrapped cells into isolation with temperature sensors at the axial ends and fitted the in-plane thermal conductivity in analytical heat equations to match recorded temperature curves. Additionally, Cheng et al. [11] cut out sections of a wounded cell stack. After sealing them inside an aluminum film, the in-plane thermal conductivity was measured with a one-dimensional temperature gradient in a sandwiched guarded hot plate setup. While all setups vary in terms of precision, flexibility and cost, no ideal approach exists.

Furthermore, Bohn et al. [12] assumed that the thermo-physical properties of lithium-ion cells including the in-plane thermal conductivity are independent of temperature. Based on measurements of dry cathode, anode and separator materials between 40 °C and 120 °C, the temperature is considered to have no effect on the in-plane thermal conductivity. However, as results of dry material measurements can not be transferred to wet cells [13] and the thermal conductivity of copper and aluminum is temperature-dependent [14], this assumption is not justified without experimental validation. Nevertheless, there does not yet exist any measurement data of the temperature-dependent in-plane thermal conductivity.

Moreover, the effect of state of charge (SOC) on the in-plane thermal conductivity was experimentally determined by Bazinski et al. [15]. Utilizing thermography and heat sensors with a one-dimensional fin equation allowed for measuring the in-plane thermal conductivity of a LFP pouch cell. Employing ten data points and two different cells, a decrease of 4.2% over the full SOC range was measured. Lin et al. [16] could not identify any dependence of the in-plane thermal conductivity of a LFP pouch cell on the SOC. Employing a thermography based measurement setup, the in-plane thermal conductivity measurements at 50% and 100% SOC did not differ significantly.

Summarizing, scattering values for the in-plane thermal conductivity of lithium-ion cells can be found in literature. Ranging from 20 $\mathrm{Wm^{-1}K^{-1}}$ to 40 $\mathrm{Wm^{-1}K^{-1}}$, only few experimental data for the in-plane thermal conductivity is available compared to extensive measurements of the through-plane thermal conductivity. While the SOC dependency of the in-plane thermal conductivity is only determined by Bazinski et al. [15] and Lin et al. [16], the influence of temperature and state of health (SOH) is unclear. In addition, the in-plane thermal conductivity is often analytically predicted by employing parallel connected thermal resistances [3]–[5]. Though, the correctness of this simulative approach is not confirmed by experimental data and lacks validation. Hence, this work presents a simple but accurate measurement approach for the in-plane thermal conductivity of lithium-ion cells. Furthermore, parametric influences on the in-plane thermal conductivity are determined and quantified. Finally, the in-plane thermal conductivity is analytically calculated by employing the thickness values of the individual cell layers and compared to experimental data. In conclusion, this work focuses on four major subjects:

- Determination of the effect of temperature on the in-plane thermal conductivity of lithium-ion cells
- Determination of the effect of SOC on the in-plane thermal conductivity of lithium-ion cells
- Determination of the effect of SOH on the in-plane thermal conductivity of lithium-ion cells
- Comparison of analytically calculated in-plane thermal conductivity values to experimental data

II. Experimental

A. Measurement setup

In order to determine the in-plane thermal conductivity, the measurement setup is based on the assembly for determining the cell cooling coefficient of Hales et al. [17]. However, several modifications to the original design are required as visible in Figure 1 (a). Aluminum bars are clamped to the opposing tabs of a pouch cell. While the free end of one aluminum bar is heated with an attached power resistor, the whole assembly is wrapped in a very low thermal conductivity rubber foam isolation with $\lambda_{\mathrm{Isolation}} = 0.038\ \mathrm{Wm^{-1}K^{-1}}$. As the setup is located inside a climate chamber and only the non-heated free end is not covered by the isolation, a temperature gradient along the sample cell can be achieved. Utilizing 27 Pt100 temperature sensors, the forced temperature delta within the cell and the aluminum bars can be precisely recorded.

B. Equivalent thermal resistance network

For calculating the in-plane thermal conductivity with the recorded temperature profile, the measurement setup is modeled as a thermal resistance network. Illustrated in Figure 1 (b), every temperature sensor in Figure 1 (a) is modeled as a node. On the one hand, $R_{\mathrm{Alu},1}$ to $R_{\mathrm{Alu},15}$ are the thermal resistances of the aluminum bars which are exactly defined since their thermal conductivity is known. On the other hand, $R_{\mathrm{Cell},1}$ to $R_{\mathrm{Cell},9}$ are the unknown thermal resistances inside the sample cell due to the unknown in-plane thermal conductivity of the cell. $R_{\mathrm{Iso-Alu}}$, $R_{\mathrm{Iso-Cell}}$ and $R_{\mathrm{Iso-Tab}}$ are thermal resistances which describe the effect of the isolation at various locations of the measurement setup. They involve a basic resistance R_{Iso} which is scaled according to the surface area that varies for the aluminum bar, the tab and the cell. Even though the thermal conductivity of the rubber foam is known, possible compression and temperature-dependent thermal behavior lead to a variable definition of R_{Iso}. Furthermore, $R_{\mathrm{Tab},1}$ and $R_{\mathrm{Tab},2}$ are the thermal resistances of the interface between the aluminum bar and the cell tab. As it is difficult to calculate an exact thermal resistance value for such a complex multi-material interface, it is considered a variable thermal resistance. The effect of convection at the non-isolated end of the aluminum bar is modeled as R_{Conv}. Since the convection coefficient can hardly be defined with satisfying accuracy, the thermal resistance is defined as a variable, too. Moreover, the ingoing heat flux $\dot{Q}_{\mathrm{in}}$ is defined as a variable as well. Because not all of the electrical power of the heater is conducted into the aluminum bar, corresponding losses are determined due to the variable definition. Lastly, the ambient temperature is implemented as the lower nodes in Figure 1 (b).

Involving the variable thermal resistances $R_{\mathrm{Tab},1}$, $R_{\mathrm{Tab},2}$, R_{Iso}, R_{Conv} and R_{Cell} and the ingoing heat flux $\dot{Q}_{\mathrm{in}}$, the problem can be formulated as an optimization problem such that the deviation between the recorded temperature gradient and the simulated temperature curve is minimized. Utilizing Matlab with the fmincon function, the following problem is solved with an interior-point algorithm:

$$\begin{aligned} \min \quad & \left|T_{\mathrm{Mes}} - T_{\mathrm{Sim}}(X)\right| \\ \textit{with} \quad & X = \left[R_{\mathrm{Tab},1}, R_{\mathrm{Tab},2}, R_{\mathrm{Iso}}, R_{\mathrm{Conv}}, R_{\mathrm{Cell}}, \dot{Q}_{\mathrm{In}}\right]^T \\ \textit{s.t.} \quad & X_{\mathrm{i}} > 0 \end{aligned} \tag{1}$$

whereby T_{Sim} is the simulated temperature data that is iteratively calculated. Therefore, equivalent resistances for the thermal resistance network in Figure 1 (b) are step by step defined

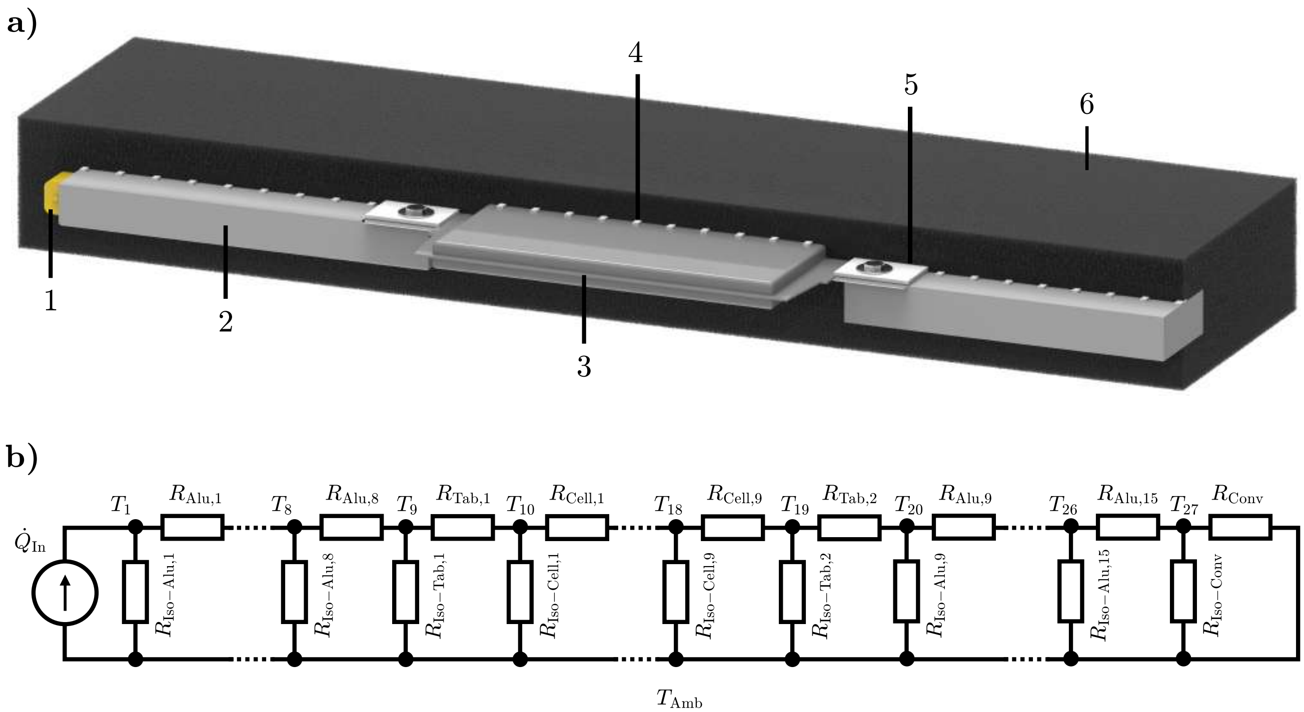

Fig. 1. Measurement setup for determining the in-plane thermal conductivity. (a) Schematic of the measurement setup with sample cell clamped between two aluminum bars. 1: Power resistor. 2: Aluminum bar. 3: Sample cell. 4: Pt100 temperature sensor. 5: Polyethylene connector for ensuring thermal contact between cell tab and aluminum bars. 6: Thermally isolating rubber foam. (b) Equivalent thermal resistance network of the measurement setup.

from the free end to the heater. Following, every temperature in Figure 1 (a) can be determined with the previously calculated equivalent resistances, the optimized ingoing heat flux as well as the heat loss through the isolation. Thus, the error between the actually measured and the simulated temperature curve is minimized within the optimization algorithm. The calculated thermal resistance of the cell R_{Cell} can be easily converted to the in-plane thermal conductivity of the cell since the height, width and the distance between the sensors are known.

TABLE I
EMPLOYED SAMPLE CELLS WITH CORRESPONDING CHARACTERISTICS.

Cell	Chemistry	Capacity	Manufacturer
#1	NCM622/Gr	60 Ah	a
#2	NCM532-NCM811/Gr	81 Ah	b
#3	NCM532-NCM811/Gr	70 Ah	b

C. Sample cells

All cells employed in this work are NCM/graphite automotive pouch cells of industrial grade. Summarized in Table I, three different cells are analyzed. While cell #1 is a begin of life 60 Ah NCM622/Gr pouch cell, cell #2 is a pristine 81 Ah NCM532-NCM811/Gr pouch cell. In contrast, cell #3 is an aged version of cell #2 which was cycled 3500 times with a 1 C charge and discharge cycle between 25% and 85% SOC at T = 24 °C such that it only has a remaining capacity of 70 Ah.

D. Validation and measurement accuracy

Since the determination of the measurement accuracy as well as the validation of the measurement approach is of high importance for the identification of parametric effects, validation measurements with a reference copper plate are performed. Therefore, a pure copper plate of 1 mm thickness is prepared such that it matches the length and width of the above described sample cells. Having a similar overall thermal resistance as common lithium-ion cells, the thermal conductivity of the copper plate is measured for varying temperatures between 10 °C and 60 °C. Corresponding results are displayed in Figure 2 (a) and can be compared to the literature value for copper obtained from Hust et al. [14].

It can be seen that the experimental data scatters between 1% and 7% from the mean value. No clear trend is observable and the scattering of the measurement results does not seem to be caused by any systematic error. Nevertheless, the mean value of 405 $\mathrm{Wm^{-1}K^{-1}}$ is very close to the literature value depicted in the graph. As the temperature-dependent decline in literature is only around 1% for the analyzed temperature range, it only has a negligible influence. Therefore, the measurement error can be specified to be below 10% which is a satisfactory accuracy when compared to other methods found in literature [18]. Furthermore, repetitive measurements are performed for the reference plate and various sample cells

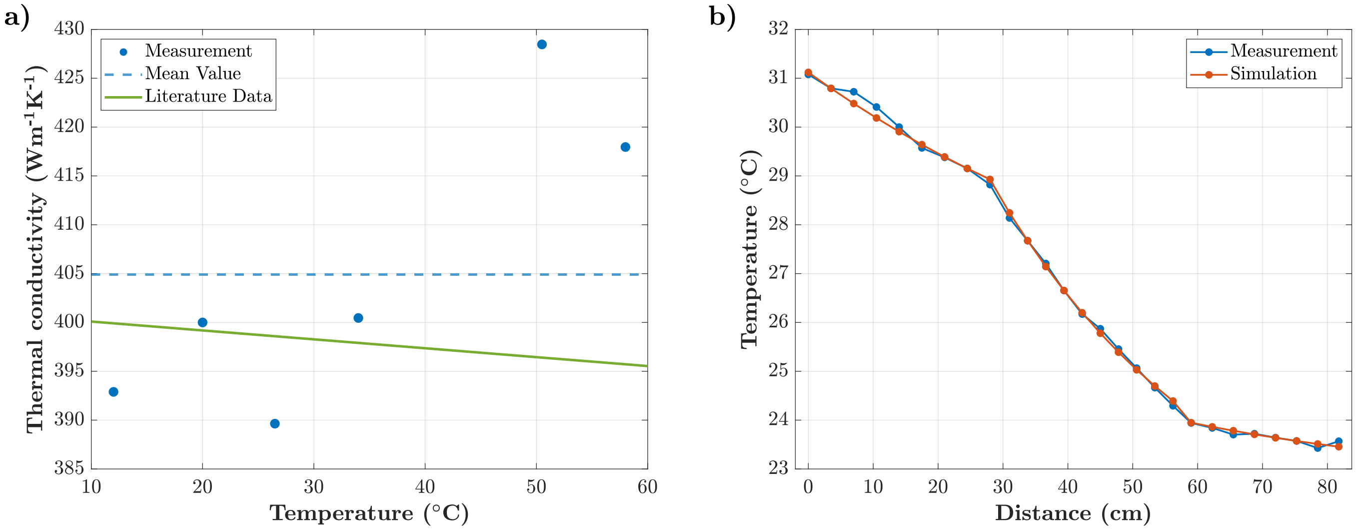

Fig. 2. Detailed results of the thermal conductivity measurements of the reference copper plate. (a) Scattering of the thermal conductivity of copper with varying temperature. (b) Comparison of measured and simulated temperature curve for the reference copper plate.

whereby the uncertainty due to reassembly does not influence the above specified measurement error of less than 10%.

Figure 2 (b) shows the results of the optimization problem for the thermal conductivity measurement of the copper plate at 26 °C. As it can be seen, the experimental data matches the simulated temperature curve well for the second aluminum bar and the copper plate. As the curves are nearly identical for the mentioned areas, only small deviations between the two temperature curves at the first aluminum bar contribute towards an average error of 0.05 K. However, the precise representation of the copper plate temperature leads to a calculated thermal conductivity of 389.6 $Wm^{-1}K^{-1}$ for the reference copper plate at 26 °C.

III. Results and Discussion

Results of the in-plane thermal conductivity measurements as well as derived parametric effects on the thermal conductivity of lithium-ion battery cells are examined in the following sections. Furthermore, found experimental results are compared to analytically simulated values. Thereby, possible explanations are validated by the analysis of thermal resistances. While the first section analyzes the influence of temperature and SOC, the second section deals with the effect of aging. Finally, a detailed analysis of thermal resistances and the analytical prediction of the in-plane thermal conductivity is given.

A. Effect of temperature and SOC on the in-plane thermal conductivity of lithium ion cells

In order to assess the influence of temperature and SOC on the in-plane thermal conductivity, cell #2 is comprehensively analyzed. Therefore, the cell is prepared as shown in Figure 1 and power cables are mounted on the cell tabs for varying the SOC. Hence, the temperature inside the climate chamber as well as the SOC can be precisely modified.

Figure 3 (a) illustrates the effect of temperature on the in-plane thermal conductivity of cell #2. While the experimental results scatter from 24 $Wm^{-1}K^{-1}$ to 26 $Wm^{-1}K^{-1}$, it is evident that the measurement uncertainty is much bigger than the scattering between the individual values. Similar results can be found for the influence of the SOC in Figure 3 (b). A comparably low scattering of the experimental data is subject to a measurement error that does not allow for the identification of any trends. In consequence, the slope of the curves in Figure 3 are caused by the measurement accuracy rather than being caused by the effect of temperature and SOC. As the effect of mentioned parametric influences is small, it can be considered negligible and the in-plane thermal conductivity can be assumed approximately constant over the analyzed operating range without significant uncertainty. Similar results are found by Werner et al. [5] who only observed a very small effect of temperature on the in-plane thermal conductivity in a thermal model. As the heat flow in the in-plane direction is expected to occur mainly along the metallic current collectors [13], the small temperature-dependent decline in thermal conductivity of less than 3% for aluminum and copper [14] is in good agreement to the non-observable effect of temperature on the in-plane thermal conductivity of the cell. Furthermore, the variation of the SOC does not influence the current collectors and therefore only affects the in-plane thermal conductivity due to the SOC-dependent thickness increase. However, since this thickness increase is only about 1% [19], the SOC has no detectable influence on the in-plane thermal conductivity of the cell, too.

B. Effect of SOH on the in-plane thermal conductivity of lithium ion cells

For determining the effect of the aging on the in-plane thermal conductivity, an aged version of cell #2 with a SOH of 86% is analyzed. While its lateral and longitudinal dimensions

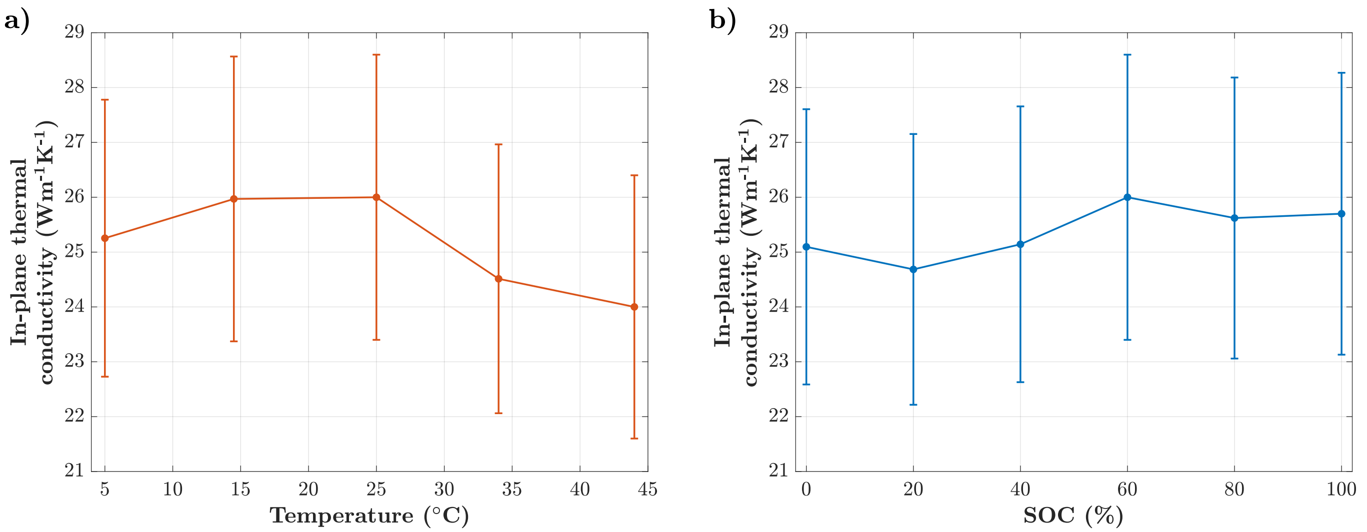

Fig. 3. Detailed results of the in-plane thermal conductivity measurements of cell #2. (a) Effect of temperature on the in-plane thermal conductivity for SOC = 60%. (b) Effect of SOC on the thermal conductivity for T = 30 °C.

remain unchanged, the thickness of cell #3 is increased by about 6% due to aging. The in-plane thermal conductivity is measured at T = 25 °C and T = 35 °C at SOC = 60% and amounts to 22.90 $Wm^{-1}K^{-1}$ and 21.56 $Wm^{-1}K^{-1}$, respectively. Thus, the aged cell has an in-plane thermal conductivity of 22.23 $Wm^{-1}K^{-1}$ on average. Compared to the pristine cell with 25.26 $Wm^{-1}K^{-1}$, there is a measurable effect of about –12% of aging on the in-plane thermal conductivity. Thus, aging seems to have a moderate influence on the in-plane thermal conductivity. However, when both measurement uncertainties are considered it is not certain whether the magnitude of this effect is actually correct. Since the copper and aluminum layers are expected to play a major role for the in-plane thermal conductivity of the cell as mentioned in the previous Section, the metallic current collectors are unaffected by aging and their thermal resistance should thus be unchanged. Because the cell thickness increases by about 6% and a decrease of the active material thermal conductivity is expected [2], a reduction of the in-plane thermal conductivity in this case of around –10% seems plausible. However, in general the magnitude of the in-plane thermal conductivity decrease strongly depends on the cell thickness increase and a more precise measurement setup is required for experimentally validating the effect of aging on the in-plane thermal conductivity in detail. Nevertheless, it can be ruled out that aging with a moderate cell thickness increase drastically influences the in-plane thermal conductivity.

C. Simulative prediction of the in-plane thermal conductivity

As thoroughly demonstrated by Ye et al. [4], it is possible to calculate an in-plane thermal conductivity of a cell with known geometric dimensions according to

$$\lambda_{\parallel} = \frac{\sum_i \lambda_i d_i}{\sum_i d_i} \tag{2}$$

whereby λ_i is the thermal conductivity of each constituting layer and d_i is the thickness of each layer. While the layer thicknesses are provided by the cell manufacturer or can be obtained through measuring a disassembled cell, the material thermal conductivities are generally obtained from literature. However, active material thermal conductivities strongly scatter [2] and thus selecting a distinct value is difficult. Since the aluminum and copper current collectors are highly thermally conductive and conduct most of the heat as proven by determining individual layer thermal resistances, the sensitivity of the in-plane thermal conductivity on the active material thermal conductivity is comparably low.

Even though several simulations employ analytically calculated in-plane thermal conductivity values, the predictive calculation via parallel thermal resistances was never compared to experimental data and thus lacks validation. As extensive geometric information as well as measurements of the in-plane thermal conductivity of all three sample cells are available, it is possible to assess the accuracy of Equation 2 first-ever. Therefore, median material thermal conductivities reported by Steinhardt et al. [2] are employed as displayed in Table II. For cell #3, a halved thermal conductivity for the anode, cathode and separator is chosen. As described in previous work [20], this is a conservative assumption. Moreover, it is assumed that the anode and cathode layer thickness increases by 5% compared to cell #2 in order to match the overall cell thickness as presented in subsection III-B. Furthermore, the measured in-plane thermal conductivity of all three cells is the average of two data points measured at T = 25 °C and T = 35 °C with an SOC of 60%.

As depicted in Table III, the resulting in-plane thermal conductivity calculated with Equation 2 for cell #1 is 19.33 $Wm^{-1}K^{-1}$ while the calculated in-plane thermal conductivity of cell #2 amounts to 23.30 $Wm^{-1}K^{-1}$. The

λ_{NCM}	$\lambda_{Graphite}$	$\lambda_{Separator}$	$\lambda_{Aluminum}$	λ_{Copper}	λ_{Pouch}
1.48 $Wm^{-1}K^{-1}$	1.04 $Wm^{-1}K^{-1}$	0.3344 $Wm^{-1}K^{-1}$	238 $Wm^{-1}K^{-1}$	398 $Wm^{-1}K^{-1}$	55.1 $Wm^{-1}K^{-1}$

TABLE II
SELECTED MEDIAN THERMAL CONDUCTIVITY VALUES FROM STEINHARDT ET AL. [2].

predicted in-plane thermal conductivity for the aged cell #3 only differs slightly from the pristine cell #2 with a value of 21.70 $Wm^{-1}K^{-1}$. Compared to the experimentally determined values, a difference of less than 10% between the experimental and calculated value exists for all three cells. Thus, analytically simulated in-plane thermal conductivities are very close to the experimentally obtained values. Especially when considering the measurement error of 10% for the in-plane thermal conductivity measurement, the inaccuracy by choosing literature thermal conductivities as well as existing uncertainties regarding the layer thickness measurements, all obtained values are within the measurement uncertainty. In consequence, it is demonstrated that the approach of parallel connected thermal resistances gives satisfying approximations of the actual in-plane thermal conductivity. When layer thicknesses and metal thermal conductivities are known, simple analytical calculations yield the approximate in-plane thermal conductivity of lithium-ion cells. Even though measurements can be performed, the simulation yields predictive in-plane thermal conductivity values which do not necessarily require experimental validation. Nevertheless, more precise measurement approaches are required to precisely quantify the accuracy of analytical models for predicting the in-plane thermal conductivity of lithium-ion cells.

TABLE III
COMPARISON OF EXPERIMENTAL DATA AND ANALYTICALLY SIMULATED IN-PLANE THERMAL CONDUCTIVITY

Cell	Measurement ($Wm^{-1}K^{-1}$)	Simulation ($Wm^{-1}K^{-1}$)
#1	20.93 ± 2.093	19.33
#2	25.26 ± 2.526	23.30
#3	22.23 ± 2.223	21.70

As the method of parallel thermal resistances is validated, the minor effect of temperature and SOC in subsection III-A can be explained by the ratio of thermal resistances within the cell. Since more than 90% of the heat is guided through the aluminum and copper layers for all three analyzed cells, only 4% of the heat flows through anode, cathode and separator. In consequence, the in-plane thermal behavior of the cell is strongly dominated by the aluminum and copper. Even though both metals show a small decline in thermal conductivity with rising temperature in the analyzed temperature range [14], the decline in in-plane thermal conductivity of less than 3% can not be recorded due to the existing measurement error of 10%. Analogously, the effect of the varying SOC does not alter the copper and aluminum of the cell. It only increases the overall cell thickness by about one percent which does not significantly affect the in-plane thermal conductivity. Therefore, a small effect of temperature and a negligible effect of SOC is confirmed. However, as the uncertainty for determining the in-plane thermal conductivity is much higher than the described parametric influences, the assumption of a constant in-plane thermal conductivity over the relevant temperature and SOC range is plausible.

Even though there exists a reasonable difference of 12% between the experimentally determined in-plane thermal conductivity for the pristine cell #2 and the aged cell #3, it can partially originate from the measurement uncertainty as explained in subsection III-B. Since both values lay within the measurement error of the analytically simulated in-plane thermal conductivity, only the analytically determined values are analyzed in the following. Therefore, the influence of aging on the in-plane thermal conductivity of cell #3 is not drastic, too. As demonstrated by the simulation in Table III, only a small decline of –7% of the in-plane thermal conductivity is observed when considering the conservative case of halved active material thermal conductivities. However, as the active material thermal conductivities only play a subordinate roll, the thickness increase due to aging is the effect which is most crucial for the alteration of the in-plane thermal conductivity. Because the overall in-plane thermal resistance stays nearly constant due to the large share of the aluminum and copper, the corresponding cross-sectional area directly correlates with the in-plane thermal conductivity. In consequence, a strong cell thickness increase can heavily influence the in-plane thermal conductivity whereby the aged active material plays a subordinate role. Hence, it is not primarily the SOH but rather the aging-related cell thickness increase which influences the in-plane thermal conductivity.

In summary, the applicability of parallel thermal resistances for the simulative prediction of the in-plane thermal conductivity of lithium-ion cells is validated. Therefore, it is demonstrated that the effects of temperature and SOC on the in-plane thermal conductivity are not significant for thermal analyses of lithium-ion cells and the assumption of a nearly constant in-plane thermal conductivity is validated. Though, aged cells have an increased cell thickness which can strongly influence the cell's in-plane thermal conductivity. The decline in active material thermal conductivity due to aging however is not of high importance for the in-plane thermal conductivity. Furthermore, the simple calculation of the heat flow through the aluminum and copper layers already gives a much more accurate in-plane thermal conductivity than the selection of highly varying literature in-plane thermal conductivity values.

IV. Conclusion

This work assessed the in-plane thermal conductivity of lithium-ion cells under varying parametric conditions. Therefore, a measurement setup that allowed for varying the temperature and SOC was developed. As the in-plane thermal conductivity is obtained through optimizing a simulated temperature curve, the measurement inaccuracy is below 10%. Most important, the experimental data matched the simulated in-plane thermal conductivity values very well. Hence, the approach of analytically predicting the in-plane thermal conductivity of lithium-ion cells was validated first-ever. Through the parallel thermal resistance analysis, it is possible to predict an accurate absolute in-plane thermal conductivity value and the parametric effects can be analytically quantified, too. As it was demonstrated that the copper and aluminum dominate the in-plane thermal behavior of the cell, the varying magnitude of the individual effects of temperature, SOC and SOH can be explained.

No significant effect of temperature and SOC on the in-plane thermal conductivity was identified. On the one hand, the experimentally recorded parametric effects of temperature and SOC for the analyzed test cells showed a lower scattering than the measurement error. On the other hand, analytical calculations confirmed the small decreasing effect on the in-plane thermal conductivity of lithium-ion cells.

Furthermore, aging decreased the in-plane thermal conductivity of the sample cell by 12%. Furthermore, it was analytically proven that a significant decrease in in-plane thermal conductivity can result from the aging-induced cell thickness increase due to the increasing cross section with a constant thermal resistance. In consequence, the influence of aging on the in-plane thermal conductivity was characterized for the first-time whereby the importance of the cell thickness increase was pointed out.

Summarizing, the parametric influences on the in-plane thermal conductivity were characterized and an analytical approach for predicting the in-plane thermal conductivity was validated. While the parametric effects were presented in detail by a thermal resistance analysis, a more precise measurement approach is required to experimentally quantify the parametric effects. Moreover, presented conclusions need to be verified by analyzing further cells with varying cell format and cell chemistry. Nevertheless, the analytical prediction gives sufficiently precise values for determining the in-plane thermal conductivity of lithium-ion cells.

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